A Genuine Bend, No Causal Certificate: Difference-in-Kinks Tests of Hyperscaler Capital Expenditure at ChatGPT

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July 2026

analysis pass of record: 2026-07, `natex` v0.2.0, seed 0

Abstract

Quarterly capital expenditure of the big-4 hyperscalers (Microsoft, Alphabet, Amazon, Meta) accelerated sharply relative to a non-AI capital-intensive comparison aggregate after the release of ChatGPT on 30 November 2022. A group difference-in-kinks (DiK) estimate at the release date is +0.1396 dex/yr (HC1 se 0.0302, nominal $p = 3.9 \times 10^{-6}$), positive in all 18 specification cells (+0.049 to +0.195), localized over 2023.0–2023.5 exactly as a lagged capex response to a late-2022 shock would be, and present per firm for Microsoft (+0.171), Alphabet (+0.206), and Amazon (+0.137) but null for Meta (+0.031, $p = 0.55$). Attribution to ChatGPT, however, fails the falsification battery: a clean pre-period placebo-cutoff grid rejects at nominal 5% in 7 of 15 non-event dates (empirical size 0.47), giving placebo-calibrated $p = 0.125$ ($|z|$) and 0.250 ($|\tau|$); the largest placebo $|z|$ (5.11, at the non-event date 2019.75) exceeds the ChatGPT estimate's matched-bandwidth $|z|$ (4.35); a full-sample kink at the non-event date 2021.5 (-0.314, $|z| = 6.9$) is larger than the main effect; and a cutoff at the BIS semiconductor export controls eight weeks earlier is statistically indistinguishable (+0.125, $z = 4.0$). Residual lag-1 autocorrelation reaches 0.66, so the nominal heteroskedasticity-robust p-values are artifacts of applying serial-independence inference to smooth aggregates. The slope divergence is a genuine, robust data feature; no causal attribution to ChatGPT survives triage.

1 Introduction

OpenAI released ChatGPT on 30 November 2022. Within two years the capital expenditure of the four large US "hyperscalers" — Microsoft, Alphabet, Amazon, and Meta — had become a macroeconomic quantity in its own right, central to debates about how much aggregate investment the current AI wave can move [1]. Financial markets repriced AI exposure almost immediately: [6] construct firm-level workforce exposures to generative AI and show that an exposure-sorted long-short portfolio earned roughly 5% in the two weeks after the release. A natural real-economy question follows: did the release bend the hyperscalers' *investment* trend — the growth rate of measured capital expenditure — or was the buildout already accelerating?

Any calendar-time answer must contend with a crowded event window. Eight weeks before ChatGPT, on 7 October 2022, the Bureau of Industry and Security announced sweeping export controls on advanced AI chips and semiconductor manufacturing equipment, a policy landmark in its own right [2]; any quarterly series treats the two events as nearly the same date.

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The estimand — a change in trend slope for one group relative to another at a known date — is the difference-in-kinks (DiK) design formalized by [4], building on the regression kink design of [5]. Kink estimates on time-series running variables are known to produce spuriously significant slope changes, which is why placebo distributions at non-event dates are the appropriate yardstick [7]; and conventional robust standard errors are badly oversized on serially correlated aggregates [3]. This paper applies the `natex` toolkit's DiK estimator [8] to a two-group quarterly capex panel built from SEC EDGAR filings, with ChatGPT as the candidate cutoff and a mandatory falsification battery — which converts a nominally overwhelming event-study result into an explicit non-attribution.

2 Data

Quarterly capital expenditure is reconstructed from SEC EDGAR XBRL `companyconcept` facts, converting fiscal year-to-date values to clean quarters by same-start differencing with remainder assignment across tag transitions. The tag is `PaymentsToAcquirePropertyPlantAndEquipment`; Amazon additionally reports under `PaymentsToAcquireProductiveAssets`. The treated series is the big-4 aggregate over 46 quarters, 2014Q4–2026Q1, rising from \$6.8bn to \$129.8bn per quarter. The rebuilt aggregate matches the frozen reference series of the run of record on all 45 common quarters (maximum difference \$0.000000bn) and its processed $\log_{10}$ counterpart on all 46 (maximum difference 5×10^{-5}).

The comparison series aggregates six large non-AI capital-intensive firms: Walmart, ExxonMobil, AT&T, Verizon, Union Pacific, and Home Depot, fetched through the identical EDGAR pipeline. Firm-specific tag quirks (AT&T, Verizon, and Home Depot report quarterly capex under `ProductiveAssets`-family tags; Walmart and Home Depot have January-31 fiscal year ends mapped to the nearest calendar quarter) are resolved against the live EDGAR records, and 2023 calendar-year sums match published figures for all six firms (AT&T \$17.85bn, Verizon \$18.77bn, ExxonMobil \$21.92bn, Walmart \$20.61bn, Union Pacific \$3.61bn, Home Depot \$3.23bn).

The outcome is $\log_{10}$ of group capex in \$bn per quarter, so slopes are in dex/yr (one dex is a factor of ten). The running variable is the quarter midpoint in fractional years; the ChatGPT cutoff is $t_0 = 2022.913$ (2022-11-30). A donut of 0.13 years ($\approx$ one quarter) removes 2022Q4, the quarter that straddles the release.

3 Design and Methods

Group difference-in-kinks. Following [4], the DiK estimand is the change in the outcome's slope in calendar time at the cutoff for the treated group *minus* the same slope change for the comparison group, estimated by local linear weighted regression on each side of the cutoff within each group (`natex` v0.2.0 [8]; the group indicator enters the CLI's DiK period dimension with a unit policy-kink change, so $\hat{\tau} = (\text{kink})_{\text{big-4}} - (\text{kink})_{\text{control}}$ in dex/yr). The primary specification uses bandwidth 3.0 years, triangular kernel, the one-quarter donut, and HC1 standard errors; $n = 46$ quarter-cells (11 pre-cutoff and 12 post-cutoff quarters per group).

Honest-inference caveats, stated as run. The estimator's own caveats apply verbatim: the cutoff and bandwidth are user-specified, so bandwidth, donut, and placebo-cutoff sensitivity must be reported; the reported interval is conventional local-polynomial inference and may retain smoothing bias; and causal interpretation requires parallel changes in non-policy slope kinks and a time-stable marginal response at the cutoff. HC1/CR1 inference additionally assumes serial independence, which these aggregates violate: residual lag-1 autocorrelation in the primary fit is 0.657 on the

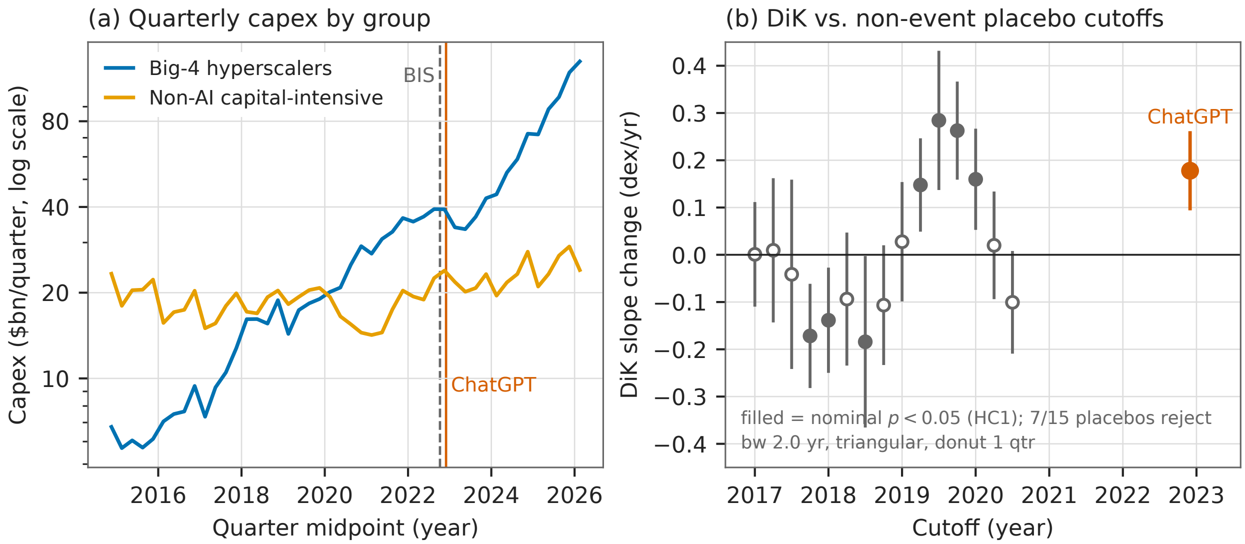

Figure 1: (a) Quarterly capital expenditure ($bn, log scale) of the big-4 hyperscaler aggregate and the non-AI capital-intensive control aggregate, 2014Q4–2026Q1, with the BIS export-control (2022-10-07, dashed) and ChatGPT (2022-11-30, solid) dates marked. (b) Group DiK estimates with 95% HC1 confidence intervals at 15 pre-period non-event placebo cutoffs (gray; pre-only sample, $t < 2022.7$) and at the ChatGPT cutoff (vermilion), all at the matched bandwidth 2.0 yr, triangular kernel, one-quarter donut; filled markers are nominally significant at the 5% level. Seven of 15 placebos reject, and the largest placebo $|z|$ exceeds the ChatGPT $|z|$.

control pre-cutoff side and 0.367 on the big-4 pre-cutoff side, precisely the setting in which nominal robust standard errors are known to be understated [3].

Falsification battery. (i) an 18-cell specification grid (bandwidth $\in \{2, 3, 4\}$ years $\times$ donut $\in \{0, 0.13, 0.25\}$ $\times$ {triangular, uniform}); (ii) a placebo cutoff at the BIS export controls (2022-10-07, $t = 2022.767$); (iii) a clean *pre-period* placebo-cutoff grid — sample restricted to $t < 2022.7$, 15 cutoffs 2017.0–2020.5, bandwidth 2.0, donut 0.13 — in the spirit of the permutation logic of [7], yielding a placebo-calibrated p-value for the main effect at the matched bandwidth; (iv) a cutoff localization scan over 2021.5–2024.75; (v) per-firm staggered DiKs of each of the ten firms against the control aggregate; and (vi) robustness checks: CR1 clustering by year and by quarter, quarter-of-year dummies, leave-one-out and no-retail control aggregates, a role-swap sanity check, and a degree-2 polynomial.

Automated pipeline, stated as run. The `natex` survey pipeline (`natex survey --seed 0`) was recorded for completeness: its automated kink family pools both groups on the auto-picked linear capex outcome and therefore cannot express the group-DiK aliasing; it returns null (Holm $p = 0.24$). The remaining families were skipped, needs-input, or failed as expected on a 92-row two-unit panel. The manual DiK runs above are the primary evidence.

4 Results

The bend is real ... The primary group DiK at ChatGPT (Table 1, first row) is $\hat{\tau} = +0.1396$ dex/yr (HC1 se 0.0302, 95% CI $[0.0804, 0.1988]$, $z = 4.62$, nominal $p = 3.85 \times 10^{-6}$). It decomposes into a big-4 kink of +0.0878 dex/yr (slope 0.0999 $\to$ 0.1876; the aggregate's growth roughly doubles, to a pace that multiplies capex by ten in about 5.3 years) and a control kink of -0.0518 (0.0811 $\to$ 0.0293). The estimate is positive in every cell of the 18-cell grid (+0.0493 to +0.1951, all nominal $p < 0.05$; the uniform kernel roughly halves $\hat{\tau}$ relative to the triangular at bandwidths ≥ 3). The point estimate is essentially invariant to CR1 clustering by year (se 0.0313) or quarter (se 0.0278), quarter-of-year dummies (+0.1396, se 0.0296), leave-one-out control aggregates (+0.1329 to +0.1515), and a no-retail control of ExxonMobil, AT&T, Verizon, and Union Pacific (+0.1453, se 0.0361); the role-swap check returns exactly -0.1396; the degree-2 estimate (+0.3022, se 0.0914) shows curvature sensitivity in magnitude but not sign. Per firm against the control aggregate: Microsoft +0.1710 (se 0.0366), Alphabet +0.2061 (se 0.0349), Amazon +0.1375 (se 0.0512), Meta +0.0308 (se 0.0518, $p = 0.55$ — a true null: Meta's surge came in 2024, after its 2022–23 pullback); control firms' own kinks are all mildly negative (-0.022 to -0.090). A localization scan (bandwidth 2.0, no donut) finds the DiK crossing zero near $t = 2022.3$, maximizing at $t = 2023.25$ (+0.2213, se 0.0440), staying above +0.19 over 2023.0–2023.75, and decaying to ≈ 0 by 2024.5 — consistent with a lagged capex response to a late-2022 shock.

... but attribution to ChatGPT fails. Three findings block a causal reading. *First*, the pre-period placebo battery fails decisively: at 15 non-event cutoffs in 2017.0–2020.5, the identical specification rejects at nominal 5% seven times (empirical size 0.467; Figure 1b). Calibrated against this placebo distribution, the ChatGPT estimate at the matched bandwidth (+0.1778, $z = 4.35$) has $p_{|z|} = (1+1)/(15+1) = 0.125$ and $p_{|\tau|} = (3+1)/16 = 0.250$; the largest placebo $|z|$ is 5.11 (at 2019.75) and the largest placebo $|\hat{\tau}|$ is 0.284 (at 2019.5) — both exceeding the ChatGPT values. *Second*, the full-sample kink at the non-event date 2021.5 is -0.3135 (se 0.0452, $|z| = 6.9$), larger in magnitude and nominal significance than the main effect: these smooth aggregates generate huge nominal kinks where nothing happened. *Third*, the cutoff is confounded: a DiK at the BIS export-control date eight weeks earlier returns +0.1253 (se 0.0317, $z = 3.95$), statistically indistinguishable from the ChatGPT estimate — quarterly data cannot separate the two events (or any other late-2022 candidate shock).

5 Caveats and Conclusion

Three caveats are decisive. First, the nominal inference is an artifact: HC1/CR1 standard errors assume serial independence, but the outcome is a pair of smooth, serially correlated aggregates (residual lag-1 autocorrelation up to 0.66), and the placebo battery measures the resulting size distortion directly — a 5% test rejects 47% of the time at non-event dates, the failure mode documented by [3] and exactly what placebo distributions exist to catch [7]. Second, a calendar-time cutoff absorbs every late-2022 shock: the BIS export controls [2] fall eight weeks before ChatGPT and produce a statistically indistinguishable estimate, so the design cannot name the shock. Third, the panel is two aggregated units observed 46 times; with ≤ 12 quarters per DiK cell, local slopes are estimated from few points on smooth curves, and the full-sample 2021.5 kink shows that this setting manufactures nominal $|z| > 6$ where nothing happened.

The verdict is descriptive-only. The post-ChatGPT divergence in capex slopes — big-4 growth roughly doubling while the control aggregate's flattens — is a genuine feature of the data: sign-robust across every specification, localized in 2023.0–2023.5 exactly as a lagged investment response to a late-2022 shock would be, and visible per firm for three of the four hyperscalers. But the design cannot certify it causally, and its headline nominal $p \approx 4 \times 10^{-6}$ overstates the evidence by roughly five orders of magnitude relative to the placebo-calibrated p of 0.125–0.25. Something bent

Table 1: Group difference-in-kinks estimates, $\log_{10}$ capex (dex/yr).

Specification	$\hat{\tau}$	se	$\lvert z \rvert$	sig. 5%
Panel A: ChatGPT cutoff ($t_0 = 2022.913$)				
triangular, bw 3, donut (primary)	+0.1396	0.0302	4.62	$*$
triangular, bw 2, donut (placebo-matched)	+0.1778	0.0409	4.35	$*$
uniform, bw 4, donut (grid minimum)	+0.0493	0.0210	2.35	$*$
primary, CR1 by year	+0.1396	0.0313	4.46	$*$
primary, quarter-of-year dummies	+0.1396	0.0296	4.72	$*$
no-retail control (XOM, T, VZ, UNP)	+0.1453	0.0361	4.03	$*$
degree 2, bw 3, donut	+0.3022	0.0914	3.30	$*$
Panel B: placebo cutoffs				
BIS export controls, $t_0 = 2022.767$, bw 3	+0.1253	0.0317	3.95	$*$
$t_0 = 2021.5$ (no event), full sample, bw 2	−0.3135	0.0452	6.93	$*$
$t_0 = 2019.75$ (no event), pre-only, bw 2	+0.2627	0.0514	5.11	$*$
$t_0 = 2019.5$ (no event), pre-only, bw 2	+0.2843	0.0736	3.86	$*$

Notes: HC1 standard errors against standard normal critical values except where noted; donut = 0.13 yr (one quarter); bandwidths in years. Outcome: $\log_{10}$ group capex, $bn per quarter. The pre-only placebo battery (sample $t < 2022.7$, 15 cutoffs 2017.0–2020.5, bw 2) rejects at nominal 5% in 7/15 cases; the placebo-calibrated p-value of the ChatGPT estimate at the matched bandwidth is 0.125 ($\lvert z \rvert$) and 0.250 ($\lvert \tau \rvert$).

hyperscaler capex around the turn of 2023; these data cannot say it was ChatGPT.

Reproducibility. All estimates were produced with `natex` v0.2.0 [8]; estimation is deterministic (local weighted least squares), and seed 0 was passed wherever the CLI accepts one. The EDGAR pulls are not committed to the repository; the run of record retains the derived panel and all estimate files. Figure 1 regenerates deterministically from the committed `figures/make_fig.py`, which asserts the headline estimates against the numbers of record before drawing.

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